

Bachelor's Thesis (Academic Year 2026)

Validation of Conservation Laws in a Quantum Vlasov Solver using a 1D Mirror Model

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Ruka Ishikawa

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Long-term kinetic simulations of the Vlasov–Maxwell system are indispensable for analyzing the self-organization and confinement physics of high- β plasmas, such as those in the RT-1 device. However, reliable evaluation of energy balance in six-dimensional phase space remains a formidable challenge for classical explicit integrators, which inevitably suffer from energy loss and non-physical dissipation over extended timescales. This study verifies the fundamental advantage of quantum algorithms not merely in terms of computational speedup, but in their capacity to ensure physical fidelity and numerical stability.

We construct a hybrid quantum–classical simulation framework for a one-dimensional magnetic mirror model, where the Vlasov time evolution is implemented via Quantum Singular Value Transformation (QSVT) to guarantee strict unitarity, while the self-consistent field is updated classically. Comparative benchmarks reveal a striking contrast: while the classical fourth-order Runge–Kutta solver exhibits catastrophic numerical divergence due to the accumulation of nonlinear errors, the quantum algorithm maintains unconditional stability with machine-precision particle number conservation and bounded energy fluctuations. These results demonstrate that the unitarity of quantum Hamiltonian simulation provides a robust and mathematically superior platform for long-term integration of fusion plasma dynamics.

Keywords: 1. Vlasov–Maxwell equations, 2. Kinetic plasma simulation, 3. Quantum Singular Value Transformation

Keio University

Faculty of Environment and Information Studies

Ruka Ishikawa

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Chapter 1

Introduction

1.1 Background

The exploration of magnetospheric plasmas has revealed fascinating phenomena, most notably the self-organization of high- β plasma in dipole magnetic fields. Experimental observations in the RT-1 device have demonstrated that plasma spontaneously forms a peaked density profile, maintaining a stable state for durations significantly exceeding the characteristic kinetic timescales [2, 3, 4]. This self-organized confinement, supported by the turbulent inward pinch [5], provides a unique laboratory for studying high-performance plasma confinement [6, 7].

However, understanding the underlying kinetic mechanisms of such long-term structural formation remains a significant challenge due to the limitations of classical numerical methods. Numerical modeling of these systems typically relies on the Vlasov–Maxwell or Vlasov–Poisson equations [8, 9]. Conventional explicit schemes, such as the fourth-order Runge-Kutta (RK4) method, are inherently non-unitary [10]. In long-term plasma simulations, these methods inevitably introduce numerical dissipation and energy loss [11]. Such artifacts can artificially decrease the system’s energy and number of particle and accumulate errors over extended integration periods [12], potentially obscuring the delicate physical balance required for self-organization.

To address these issues, the paradigm of quantum simulation offers a fundamentally different approach [13, 14]. Quantum algorithms for the Vlasov equation [15] utilize the principle of Hamiltonian simulation to evolve the system. Recent advances in Quantum Signal Processing (QSP) [16] and Quantum Singular Value Transformation (QSVT) [17, 18] have enabled the implementation of such simulations with nearly optimal dependence on parameters.

While quantum algorithms are often discussed in terms of their potential to improve computational complexity [19, 1], the primary motivation of this study lies in their inherent numerical accuracy and stability. Quantum Hamiltonian simulation evolves the system via unitary operators, which correspond to rotations in a complex vector space [20]. This unitarity ensures the strict conservation of the number of particles.

Crucially, because the time evolution is strictly unitary, quantum solvers can principally suppress energy loss, a feat that is difficult to achieve with non-unitary classical integrators without significant computational overhead. Building on the work by Higuchi et al. [21], who evaluated Hamiltonian simulation for instability benchmarks, this study aims to verify the long-term conservation laws and numerical stability of quantum solvers in the context of magnetic mirror confinement. By leveraging the exactness of unitary evolution, quantum-classical hybrid approaches provide a robust computational framework required for investigating the structural formation of magnetospheric plasmas without the interference of numerical dissipation.

1.2 Contributions

This study establishes a foundational framework for verifying the numerical stability of quantum kinetic plasma simulations, integrating physical modeling of magnetic confinement with rigorous conservation analysis. The primary contributions of this research are summarized as follows:

1. **Application of Quantum Hamiltonian Simulation to Magnetic Mirror Confinement**

We extended the application of quantum algorithms from simple periodic systems to a bounded magnetic mirror configuration, which models the fundamental dynamics along a magnetic field line in dipole plasmas like the RT-1 device. Unlike previous studies that focused on linear advection or instability growth in periodic domains, this work demonstrates the applicability of quantum time evolution to trapped particle dynamics in a non-uniform external potential.

2. **Validation of Unitary Time Evolution using QSVT**

We implemented the time evolution of the Vlasov equation as a unitary operator using Quantum Singular Value Transformation (QSVT). By semi-discretizing the phase space with central differences to construct a skew-Hermitian Liouvillean, we mapped the dynamics onto a Hermitian Hamiltonian suitable for quantum simulation. This formulation ensures that the time evolution operator is strictly unitary, preserving the quantum state norm throughout the computation.

3. **Demonstration of Numerical Stability and Conservation Laws**

A critical contribution of this study is the quantitative verification of conservation laws in a quantum-classical hybrid setting. We demonstrated that the quantum solver strictly preserves the total particle number and suppresses energy loss over long integration periods. This stands in sharp contrast to classical explicit schemes, such as the fourth-order Runge-Kutta method, which exhibited significant dissipation and energy drift under identical conditions.

4. Establishment of a Robust Framework for Long-Term Plasma Simulations

We confirmed that the unitarity of quantum algorithms provides a distinct advantage for long-term simulations required to study self-organization phenomena. By effectively eliminating numerical diffusion without the complex implementation overhead of classical symplectic or semi-Lagrangian schemes, this research positions quantum-classical hybrid solvers as a promising approach for future high-fidelity modeling of collisionless plasmas.

1.3 Thesis Structure

The remainder of this thesis is organized as follows. Chapter 2 provides the preliminary knowledge required for this study, covering the fundamentals of quantum computation, the Vlasov-Maxwell system, the classical numerical simulation, and the QSVT algorithm. Chapter 3 defines the specific problem addressed in this research. Chapter 4 details the methodology of the proposed hybrid simulation model, including the Hamiltonian block encoding and the classical-quantum data coupling scheme. Chapter 5 presents the numerical simulation results and analyzes Numerical diffusion and energy loss. Finally, Chapter 6 summarizes the conclusions.

Chapter 2

Preliminaries

2.1 Mathematical Notations and Definitions

In this thesis, we utilize standard linear algebra and quantum mechanical notations. We explicitly define the key terms and operators used throughout the subsequent chapters as follows:

Unitary Operator : A linear operator U is unitary if its inverse equals its adjoint, i.e., $U^\dagger U = U U^\dagger = I$, where I is the identity operator. Unitary operators preserve the L^2 -norm of a vector, which is crucial for describing the time evolution of closed quantum systems and ensuring particle number conservation in our Vlasov simulation.

Hamiltonian: In the context of quantum mechanics, the Hamiltonian H is a Hermitian operator ($H = H^\dagger$) representing the total energy of the system. It generates the time evolution of a quantum state via the Schrödinger equation. Note that in this study, we map the discretized Liouvillian of the Vlasov equation to an effective Hamiltonian matrix for simulation on a quantum computer.

Projector : A projector (or projection operator) P is a Hermitian operator that satisfies $P^2 = P$. In the context of Block Encoding, a projector is used to specify the subspace where the target matrix is encoded. Typically, we use $P = |0\rangle\langle 0| \otimes I$ to project onto the ancilla-zero state.

Singular Values : For any matrix A , the singular values σ_i are the square roots of the eigenvalues of $A^\dagger A$. In the Quantum Singular Value Transformation (QSVT) algorithm, we perform polynomial transformations on these singular values to implement functions of the matrix A .

Operator Norm : The spectral norm (or operator norm) $\|A\|$ is defined as the largest singular value of A , i.e., $\|A\| = \max_{x \neq 0} \frac{\|Ax\|}{\|x\|}$. In Hamiltonian simulation,

the matrix must be scaled such that its operator norm is less than or equal to 1 ($\|A\| \leq 1$) to be embedded into a unitary operator.

2.2 Quantum Computers

Quantum computation comprises new computational models that process information based on the fundamental principles of quantum mechanics. By utilizing quantum phenomena such as superposition and entanglement, they have the potential to provide fast and efficient solutions to problems that are inefficient or infeasible for classical computers [13, 14]. Since the introduction of Shor’s factorization algorithm and Grover’s search algorithm in the 1990s, the theoretical superiority of quantum computation has been demonstrated, leading to rapid advancements in both theory and implementation. This section provides an overview of the basic components of quantum computation, including qubits and quantum circuits, largely based on standard texts [1].

2.2.1 Qubits

The “bit” is the smallest unit of information in classical computation, which can take either 0 or 1. In quantum computation, the basic unit is called a “quantum bit” or “qubit.” A qubit is represented as a two-dimensional vector in Hilbert space and can be expressed as a linear combination of the basis states $|0\rangle$ and $|1\rangle$:

$$|\psi\rangle = \alpha |0\rangle + \beta |1\rangle, \quad \alpha, \beta \in \mathbb{C}, \quad |\alpha|^2 + |\beta|^2 = 1 \quad (2.2.1)$$

Here, α and β are complex amplitudes, and the squared magnitudes $|\alpha|^2$ and $|\beta|^2$ correspond to the probabilities of measuring the states $|0\rangle$ and $|1\rangle$, respectively. Such a superposition state is a uniquely quantum concept with no classical counterpart, and it forms the foundation for the inherent parallelism in quantum algorithms.

Furthermore, multiple qubits can form nonlocal correlated states known as *entanglement*, enabling an exponential expansion of the accessible state space. A system of n qubits belongs to the 2^n -dimensional complex Hilbert space $\mathbb{C}^{2^n}$, and its general state can be expressed as:

$$|\psi\rangle = \sum_{j=0}^{2^n-1} \alpha_j |j\rangle, \quad \sum_j |\alpha_j|^2 = 1 \quad (2.2.2)$$

Describing such a state classically requires memory resources that grow exponentially with n , highlighting the expressive power of quantum information.

In addition, the state of a single qubit can be visualized geometrically as a point on a three-dimensional sphere known as the *Bloch sphere*. This representation reflects the fact that a one-qubit pure state belongs to the complex projective space $\mathbb{CP}^1$, and

provides an intuitive framework for understanding unitary rotations and measurement outcomes.

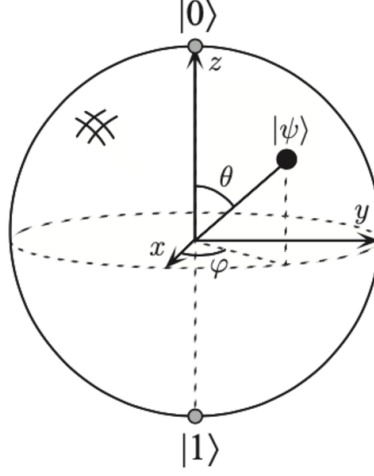


Figure 2.1: Bloch sphere [1].

2.2.2 Quantum Gates

In quantum computation, operations are implemented via unitary operators, analogous to time evolution in quantum mechanics. In the quantum circuit model, unitary transformations are constructed by sequentially applying basic quantum gates to qubits. These include single-qubit gates such as the Hadamard, Pauli, and phase gates, as well as multi-qubit gates like the CNOT gate. This construction mirrors the way logic gates are composed in classical circuits [1].

For example, the Hadamard gate H is represented by the matrix

$$H = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}, \quad (2.2.3)$$

which transforms the basis state $|0\rangle$ into the superposition state $(|0\rangle + |1\rangle)/\sqrt{2}$.

Other representative single-qubit gates include the Pauli gates:

$$X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad Y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad (2.2.4)$$

as well as rotation gates around different axes, $R_X(\theta)$, $R_Y(\theta)$, $R_Z(\theta)$, and the phase gate:

$$P(\phi) = \begin{pmatrix} 1 & 0 \\ 0 & e^{i\phi} \end{pmatrix}. \quad (2.2.5)$$

Among multi-qubit gates, the controlled-NOT (CNOT) gate plays a crucial role. It applies an X operation to the target qubit conditioned on the control qubit being in the state $|1\rangle$. The CNOT gate is essential for generating entanglement and for quantum error correction. When combined with single-qubit gates, it forms a universal gate set capable of realizing any unitary operation on multiple qubits.

Representative two-qubit gates are shown below (ordered in the computational basis $|00\rangle, |01\rangle, |10\rangle, |11\rangle$):

- **CNOT gate:** Applies an X gate to the target qubit if the control qubit is in state $|1\rangle$:

$$\text{CNOT} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix} \quad (2.2.6)$$

- **SWAP gate:** Exchanges the states of two qubits:

$$\text{SWAP} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad (2.2.7)$$

- **CZ gate** (controlled-Z): Applies a Z gate to the target qubit if the control qubit is in state $|1\rangle$:

$$\text{CZ} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 \end{bmatrix} \quad (2.2.8)$$

The Toffoli gate (also known as the controlled-controlled-NOT or CCNOT gate) operates on three qubits. It flips the target qubit if both control qubits are in the state $|1\rangle$. Its matrix representation in the basis $|000\rangle, |001\rangle, |010\rangle, |011\rangle, |100\rangle, |101\rangle, |110\rangle, |111\rangle$ is given by:

- **Toffoli gate:**

$$\text{Toffoli} = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \end{bmatrix} \quad (2.2.9)$$

2.2.3 Measurement

In quantum computation, the final output is obtained through a process called *measurement*. This operation projects a quantum state—represented as a linear combination of basis vectors with complex amplitudes—onto a specific basis, typically the computational basis. Upon measurement, the quantum state undergoes a probabilistic *collapse* to one of the basis states, resulting in a classical bit string as output. The probability of obtaining a specific outcome corresponds to the squared modulus of the amplitude associated with that basis state.

For example, consider a single-qubit state

$$|\psi\rangle = \alpha |0\rangle + \beta |1\rangle. \quad (2.2.10)$$

When measured in the computational basis $\{|0\rangle, |1\rangle\}$, the outcome 0 occurs with probability $|\alpha|^2$, and the outcome 1 with probability $|\beta|^2$. Immediately after the measurement, the state collapses irreversibly to the corresponding basis vector, i.e., either $|0\rangle$ or $|1\rangle$.

This irreversible transformation plays a crucial role in the design of quantum algorithms. The timing and nature of measurement within a quantum circuit can significantly affect the overall efficiency and success probability of the algorithm [1]. Quantum algorithms typically follow a three-stage structure: *state preparation* $\rightarrow$ *unitary operations* $\rightarrow$ *measurement*. Since measurement is the only way to extract information from a quantum system into classical form, designing efficient strategies to retrieve the required data—such as phase, amplitude, or the value of specific bits—with a finite number of measurements is a critical aspect of quantum algorithm design.

2.2.4 Quantum Oracles

A quantum oracle is a fundamental subroutine in quantum algorithms, acting as a “black box” unitary operation that provides access to specific data or functions. In the context of quantum simulation, oracles are essential for encoding the classical information of the Hamiltonian matrix into the quantum circuit [19, 20].

Mathematically, an oracle O acts on computational basis states. A standard binary oracle for a function $f(x)$ typically operates as:

$$O |x\rangle |y\rangle = |x\rangle |y \oplus f(x)\rangle, \quad (2.2.11)$$

where $\oplus$ denotes addition modulo 2. In this study, specifically for the Block Encoding of the Vlasov evolution operator, we utilize sparse matrix oracles. These oracles query the row indices and matrix element values of the discretized Hamiltonian H . By constructing such oracles, the classical data of the differential operators is embedded into the quantum processor, allowing algorithms like Quantum Singular Value Transformation (QSVT) to process the dynamics without explicitly storing the full matrix in classical memory [17, 18, 15].

2.2.5 Quantum Amplitude Estimation (QAE)

While quantum simulation evolves the state vector to represent the physical distribution function $f(x, v, t)$, extracting useful physical quantities (observables) from the quantum state requires efficient measurement strategies. Quantum Amplitude Estimation (QAE) is a celebrated algorithm that estimates the amplitude a of a target state with a quadratic speedup over classical sampling methods [1].

Given a quantum algorithm $\mathcal{A}$ such that $\mathcal{A} |0\rangle = \sqrt{a} |\psi_{\text{good}}\rangle + \sqrt{1-a} |\psi_{\text{bad}}\rangle$, QAE estimates the probability a with an accuracy of ϵ using $O(1/\epsilon)$ queries to $\mathcal{A}$. This is a significant improvement compared to the $O(1/\epsilon^2)$ scaling of classical Monte Carlo integration. In kinetic plasma simulations, QAE is crucial for computing moments of the distribution function, such as particle density $n(x) = \int f dv$ or total energy, from the final quantum state [21, 15]. It allows for the readout of macroscopic physical quantities with high precision, bridging the gap between the quantum state vector and experimentally comparable values.

2.3 Hamiltonian Simulation

One of the most prominent applications of quantum computing is *Hamiltonian simulation*, which involves emulating the time evolution of quantum systems. This problem arises naturally in physics, chemistry, materials science, and even in fields such as plasma physics and quantum field theory. In the context of plasma fusion research, understanding the behavior of high-beta plasmas and self-organized confinement—as explored in the RT-1 device [2, 6, 3, 7, 4] and levitated dipoles [5]—is crucial. While classical simulation methods like Particle-in-Cell (PIC) and Boris algorithms have been successful [11, 12], quantum computing offers an intrinsic computational advantage for solving complex kinetic equations such as the Vlasov equation [13, 14, 15, 21]. For a closed quantum system governed by a time-independent Hamiltonian H , the evolution of the quantum state $|\psi(t)\rangle$ is described by the time-dependent Schrödinger equation:

$$i \frac{d}{dt} |\psi(t)\rangle = H |\psi(t)\rangle. \quad (2.3.1)$$

When the Hamiltonian H is time-independent, the formal solution to the Schrödinger equation is given by

$$|\psi(t)\rangle = e^{-iHt} |\psi(0)\rangle. \quad (2.3.2)$$

Therefore, Hamiltonian simulation on a quantum computer aims to approximate this time-evolution operator e^{-iHt} using a quantum circuit [1]. By applying it to an initial state $|\psi(0)\rangle$, the simulation reproduces the physical time dynamics of the system.

2.3.1 Trotter Decomposition

One of the earliest approaches to Hamiltonian simulation is the *Trotter decomposition* [14]. When the Hamiltonian can be expressed as a sum of local terms, $H = \sum_{j=1}^L H_j$, the time evolution operator can be approximated by the product of exponentials of each term. The first-order Trotter formula takes the form:

$$e^{-iHt} \approx \left(\prod_{j=1}^L e^{-iH_j t/r} \right)^r + \mathcal{O}(t^2/r), \quad (2.3.3)$$

where r is the number of Trotter steps. To guarantee a precision ε , one typically needs $r = \mathcal{O}(t^2/\varepsilon)$. Higher-order schemes such as Suzuki decompositions can improve the accuracy per step, but they often incur significant circuit depth and gate complexity, which poses challenges for implementation on NISQ devices.

While Trotterization is intuitive and often exploits the structure of the physical system, it is not optimal from a complexity perspective. This has led to the development of more advanced approaches such as the *Linear Combination of Unitaries* (LCU) method [19] and *Qubitization* [20], which avoid direct decomposition of exponentials and provide theoretically optimal constructions.

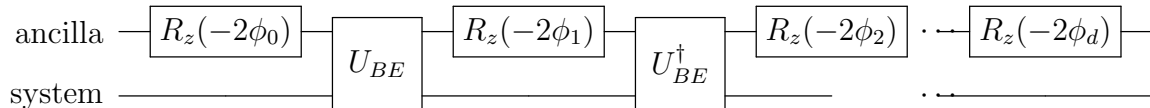
2.3.2 Quantum Singular Value Transformation

Quantum Singular Value Transformation (QSVT) is a powerful quantum algorithmic framework that allows one to implement polynomial transformations on the singular values of a block-encoded matrix [17, 18]. This general technique subsumes many existing quantum algorithms, including Hamiltonian simulation, amplitude amplification, and quantum linear system solvers.

Let a matrix A be block-encoded as $A = \tilde{\Pi}U\Pi$, where U is a unitary operator and $\Pi, \tilde{\Pi}$ are projectors onto appropriate subspaces. Given a polynomial P and a phase vector $\varphi = (\phi_0, \phi_1, \dots, \phi_d) \in \mathbb{R}^{d+1}$, one constructs a unitary U_φ as

$$U_\varphi = \begin{cases} e^{i\phi_0(2\tilde{\Pi}-I)}U \prod_{j=1}^{(d-1)/2} \left(e^{i\phi_{2j}(2\Pi-I)}U^\dagger e^{i\phi_{2j+1}(2\tilde{\Pi}-I)}U \right), & \text{if } d \text{ is odd,} \\ \prod_{j=1}^{d/2} \left(e^{i\phi_{2j-1}(2\Pi-I)}U^\dagger e^{i\phi_{2j}(2\tilde{\Pi}-I)}U \right), & \text{if } d \text{ is even.} \end{cases} \quad (2.3.4)$$

The structure of the QSVT circuit, characterized by alternating applications of the block-encoded unitaries U_{BE} (and $U_{BE}^\dagger$) and phase rotations R_z , is visualized below:



The resulting transformed operator satisfies:

$$\tilde{\Pi}U_\varphi\Pi = \begin{cases} \sum_{k=1}^r P(\sigma_k)|w_k\rangle\langle v_k|, & \text{if } d \text{ is odd,} \\ \sum_{k=1}^r P(\sigma_k)|v_k\rangle\langle v_k|, & \text{if } d \text{ is even,} \end{cases} \quad (2.3.5)$$

where σ_k are the singular values of A and $P(\cdot)$ is a polynomial constructed via Quantum Signal Processing (QSP).

To ensure unitarity and implementability, $P(x)$ must satisfy the following conditions:

- (i) P has parity $d \bmod 2$,
- (ii) $\forall x \in [-1, 1], |P(x)| \leq 1$,
- (iii) $\forall x \in (-\infty, -1] \cup [1, \infty), |P(x)| \geq 1$,
- (iv) If d is even, then $\forall x \in \mathbb{R}, P(ix)P^*(ix) \geq 1$.

When only the real part of P is used, conditions (iii) and (iv) can be relaxed.

2.3.3 Block-Encoding

Block-encoding is a standard method in quantum computing for embedding a non-unitary matrix $\hat{H}$ into a larger unitary matrix U . Specifically, we say that a unitary U is an $(\eta\alpha, a, 0)$ -block encoding of $\hat{H}$ if

$$U = \begin{pmatrix} \hat{H}/(\eta\alpha) & \cdot \\ \cdot & \cdot \end{pmatrix}, \quad (2.3.6)$$

where η is a normalization factor arising from the structure of ancilla qubits, α is a known scaling factor, and a is the number of ancilla qubits used.

For a specific Hamiltonian H with normalization $\alpha = \|H\|$, the block encoding unitary U_{BE} is often constructed to satisfy:

$$U_{BE} = \begin{pmatrix} H/\alpha & \sqrt{I - (H/\alpha)^2} \\ \sqrt{I - (H/\alpha)^2} & -H/\alpha \end{pmatrix}$$

where the top-left block corresponds to the projection $\langle 0|U_{BE}|0\rangle = H/\alpha$.

Block-encoding allows for matrix operations (addition, multiplication, etc.) to be simulated within quantum algorithms [17]. If $\|\hat{H}\| \geq 1$, the normalization factor ensures that the encoded operator has bounded norm and the entire structure remains unitary.

In applications such as Hamiltonian simulation, the time evolution operator $e^{-i\hat{H}\Delta t}$ can be approximated by applying QSVT to the block-encoded $\hat{H}$. A time scaling parameter τ is introduced such that $\Delta t = \tau/(\eta\alpha)$, and the approximation is constructed using polynomial approximations to the cosine and sine functions.

2.3.4 Quantum Signal Processing

Quantum Signal Processing (QSP) is a technique that constructs unitary operators whose entries implement polynomial transformations of a real variable $x \in [-1, 1]$, which often corresponds to eigenvalues or singular values of some underlying matrix [16].

The QSP circuit applies alternating sequences of phase shifts and signal-dependent rotations. For a given phase sequence $\varphi = (\phi_0, \phi_1, \dots, \phi_d)$, the QSP unitary is constructed as:

$$U_\varphi(x) = e^{i\phi_0\sigma_z} R(x) e^{i\phi_1\sigma_z} R(x) \cdots R(x) e^{i\phi_d\sigma_z}, \quad (2.3.7)$$

where $R(x)$ is a reflection operator defined by

$$R(x) = \begin{bmatrix} x & \sqrt{1-x^2} \\ \sqrt{1-x^2} & -x \end{bmatrix}. \quad (2.3.8)$$

QSP enables the realization of real or complex polynomials $P(x)$ satisfying:

- P has degree d ,
- P is either even or odd,
- $|P(x)| \leq 1$ on $x \in [-1, 1]$.

For example, Chebyshev polynomials $T_d(x)$ can be exactly implemented by QSP with a known choice of φ . More generally, for any analytic function $f(x)$, one can construct a polynomial approximation $P(x) \approx f(x)$, and then synthesize a corresponding QSP sequence.

Combined with block-encoding, QSP forms the computational backbone of QSVT. It enables efficient and high-precision implementations of spectral transformations such as e^{-iHt} , $\text{sign}(H)$, or matrix inversion H^{-1} , using only a small number of ancilla qubits and queries to the underlying unitary.

2.3.5 Simulation Strategy via LCU and QSVT

To simulate the time evolution operator $e^{-iH\tau}$, one effectively utilizes Euler's formula $e^{-iH\tau} = \cos(H\tau) - i \sin(H\tau)$. Since QSVT naturally implements polynomials of definite parity, the even part ($\cos$) and the odd part ($-i \sin$) are approximated separately by two distinct QSVT sequences, $U_{\cos}$ and $U_{\sin}$.

These components are then combined using the Linear Combination of Unitaries (LCU) technique [19]. An ancilla qubit is used to control the application of the real and imaginary parts. The detailed circuit structure is shown below:

One of the defining features of plasma is its strong interaction with electromagnetic fields, leading to rich dynamical phenomena such as Larmor motion, drift motion, wave propagation, magnetic reconnection, and various instabilities. Describing plasma dynamics requires a statistical and electromagnetic framework rather than simple fluid dynamics. Consequently, plasma physics often relies on kinetic approaches, notably the Vlasov equation for the evolution of particle distribution functions, coupled with Maxwell's equations governing electromagnetic fields.

Understanding and controlling plasma behavior is crucial for the advancement of astrophysics and fusion energy research, as plasma encompasses a wide variety of complex and dynamic physical processes.

2.4.2 Distribution Functions and Their Physical Meaning

Due to the enormous number of particles in a plasma, it is impractical to track the motion of each individual particle. Instead, a statistical description is adopted, centered around the concept of a distribution function. The distribution function $f(\mathbf{x}, \mathbf{v}, t)$ represents the probability density of particles in phase space, such that $f(\mathbf{x}, \mathbf{v}, t) d^3\mathbf{x} d^3\mathbf{v}$ gives the expected number of particles located near position $\mathbf{x}$ with velocity $\mathbf{v}$ at time t [9].

Defined in the six-dimensional phase space $(\mathbf{x}, \mathbf{v})$, this function characterizes the kinetic behavior of electrons, ions, and other plasma constituents. Macroscopic quantities such as number density, mean velocity, and energy density can be derived as moments of the distribution function over velocity space. For instance, the number density is given by

$$n(\mathbf{x}, t) = \int_{\mathbb{R}^3} f(\mathbf{x}, \mathbf{v}, t) d^3\mathbf{v}, \quad (2.4.1)$$

and the mean velocity is

$$\mathbf{u}(\mathbf{x}, t) = \frac{1}{n(\mathbf{x}, t)} \int_{\mathbb{R}^3} \mathbf{v} f(\mathbf{x}, \mathbf{v}, t) d^3\mathbf{v}. \quad (2.4.2)$$

Thus, the distribution function serves as a bridge between microscopic particle dynamics and macroscopic observables. Especially under non-equilibrium conditions or in the presence of strong electromagnetic fields, it provides a more accurate description than fluid models. Its evolution is governed by the Vlasov equation, which, together with Maxwell's equations, forms a self-consistent framework for modeling plasma dynamics.

2.4.3 Derivation of the Vlasov Equation from Liouville's Theorem

In classical statistical mechanics, the evolution of a distribution function in phase space is governed by Liouville's theorem, which asserts the conservation of phase

space density along Hamiltonian flows. We begin by deriving this theorem and then proceed to apply it to derive the Vlasov equation, which governs the dynamics of collisionless plasmas [8].

Consider a system of particles evolving in a 6-dimensional phase space $\Gamma = \{\mathbf{x}, \mathbf{v}\}$, where $\mathbf{x} \in \mathbb{R}^3$ denotes the spatial coordinates and $\mathbf{v} \in \mathbb{R}^3$ denotes the velocity coordinates. Let $f(\mathbf{x}, \mathbf{v}, t)$ be the single-particle distribution function, representing the probability density in phase space.

The total time derivative of f along a trajectory in phase space is given by

$$\frac{df}{dt} = \frac{\partial f}{\partial t} + \frac{d\mathbf{x}}{dt} \cdot \nabla_{\mathbf{x}} f + \frac{d\mathbf{v}}{dt} \cdot \nabla_{\mathbf{v}} f. \quad (2.4.3)$$

Let us consider the time evolution of an infinitesimal volume element in phase space, $d\Gamma = d^3\mathbf{x} d^3\mathbf{v}$. Under the Hamiltonian flow generated by the equations of motion:

$$\frac{d\mathbf{x}}{dt} = \mathbf{v}, \quad (2.4.4)$$

$$\frac{d\mathbf{v}}{dt} = \frac{\mathbf{F}}{m}, \quad (2.4.5)$$

the time evolution of this volume is given by the Jacobian determinant of the transformation. Liouville's theorem states that this Jacobian determinant is unity, implying that the phase-space volume is conserved, and hence the distribution function is constant along trajectories:

$$\frac{df}{dt} = 0. \quad (2.4.6)$$

We now apply Liouville's theorem to the dynamics of a plasma composed of charged particles under electromagnetic fields. Let q and m denote the charge and mass of each particle, respectively. The motion of particles is governed by the Lorentz force:

$$\frac{d\mathbf{x}}{dt} = \mathbf{v}, \quad (2.4.7)$$

$$\frac{d\mathbf{v}}{dt} = \frac{q}{m} (\mathbf{E}(\mathbf{x}, t) + \mathbf{v} \times \mathbf{B}(\mathbf{x}, t)). \quad (2.4.8)$$

Substituting into Eq. (2.4.3) and applying Eq. (2.4.6), we arrive at the collisionless Vlasov equation:

$$\frac{\partial f}{\partial t} + \mathbf{v} \cdot \nabla_{\mathbf{x}} f + \frac{q}{m} (\mathbf{E} + \mathbf{v} \times \mathbf{B}) \cdot \nabla_{\mathbf{v}} f = 0. \quad (2.4.9)$$

Equation (2.4.9) describes the time evolution of the distribution function in the absence of collisions. It is valid for high-temperature, low-density plasmas where the mean free path is much larger than the characteristic system size.

2.4.4 Coupling with Maxwell's Equations

The electromagnetic fields in a plasma are generated by the motion of its charged particles and simultaneously influence that motion. To capture this mutual interaction, the Vlasov and Maxwell equations must be coupled.

The charge and current densities derived from the distribution function are given by

$$\rho(\mathbf{x}, t) = q \int f(\mathbf{x}, \mathbf{v}, t) d^3\mathbf{v}, \quad \mathbf{J}(\mathbf{x}, t) = q \int \mathbf{v} f(\mathbf{x}, \mathbf{v}, t) d^3\mathbf{v}. \quad (2.4.10)$$

Incorporating these into Maxwell's equations yields a self-consistent set of equations known as the Vlasov–Maxwell system [11]. This coupled framework is central to applications in fusion, space plasma, and beam physics.

2.4.5 Hamiltonian Structure of the Vlasov–Maxwell System

Under idealized conditions with no collisions or dissipation, the Vlasov–Maxwell system conserves total energy and exhibits a Hamiltonian structure. This perspective enables a unified understanding of its dynamics in terms of functional derivatives and noncanonical Poisson brackets.

When the distribution function f is viewed as a functional over phase space, the evolution of any observable F is given by

$$\frac{dF}{dt} = \{F, H\}, \quad (2.4.11)$$

where H denotes the system's Hamiltonian and $\{\cdot, \cdot\}$ represents the noncanonical Poisson bracket. This formulation explicitly captures the non-commutative nature of interacting observables.

Furthermore, the Hamiltonian structure is of critical importance in the context of quantum algorithms. In particular, it ensures unitary time evolution under operators such as $\exp(-iHt)$, making the Vlasov–Maxwell system well-suited for advanced simulation techniques like QSVT-based Hamiltonian simulation [19, 17, 15].

2.4.6 Numerical Simulation Methods in Plasma Physics

Plasma is a complex multi-body system where charged particles and electromagnetic fields interact self-consistently. Various numerical simulation methods have been developed depending on the target scales and physical processes [11, 10]. The representative approaches are the Particle-In-Cell (PIC) method, the Magnetohydrodynamics (MHD) model, and the Vlasov simulation.

Particle-In-Cell (PIC) Method

The PIC method is a numerical technique that directly tracks the trajectories of a large number of charged particles (macro-particles) while solving the electromagnetic fields on a spatial grid [11]. Particles evolve according to the Lorentz force, and their charge and current densities are interpolated onto grid points to update the electromagnetic fields via Maxwell’s equations.

This approach naturally incorporates kinetic effects, allowing for the description of wave-particle interactions and nonlinear phenomena. However, since it employs a finite number of particles, statistical noise is unavoidable.

Magnetohydrodynamics (MHD) Model

The MHD model treats plasma as a conducting fluid continuum, evolving macroscopic physical quantities such as density, velocity, pressure, and magnetic fields [8]. The primary advantage of the MHD model is its relatively low computational cost, making it suitable for describing large-scale and long-duration phenomena. Nevertheless, it possesses a fundamental limitation: it cannot account for kinetic effects such as velocity-space structures.

Vlasov Simulation (Distribution Function Method)

Vlasov simulation is a method that directly evolves the distribution function $f(\mathbf{x}, \mathbf{v}, t)$ in phase space rather than tracking individual particles. By solving the Vlasov equation coupled with Maxwell’s equations, it provides a noise-free kinetic description of the plasma.

This method excels in capturing fine structures in velocity space. However, because the phase space is high-dimensional (reaching six dimensions in a 3D3V configuration), the number of required grid points grows exponentially. This leads to the “curse of dimensionality,” presenting a fatal challenge for classical computers [15].

Comparison of Methods and Positioning of This Study

This study focuses on the high computational cost of Vlasov simulations and explores the possibility of implementing the time evolution of the distribution function using quantum computing algorithms. Specifically, by applying the Quantum Singular Value Transformation (QSVT) [17, 18] to the effective Hamiltonian derived from the semi-discretization of the Vlasov equation, we aim to establish a foundation for a novel computational methodology for kinetic plasma simulations [15, 21].

2.4.7 General Principles of Magnetic Confinement

The fundamental concept of magnetic confinement fusion is to contain a high-temperature plasma within a vacuum vessel by utilizing the interaction between charged particles

and a magnetic field. The motion of a charged particle with charge q and velocity $\mathbf{v}$ in a magnetic field $\mathbf{B}$ is governed by the Lorentz force $\mathbf{F} = q(\mathbf{v} \times \mathbf{B})$. This force acts perpendicular to both the velocity and the magnetic field, causing the particle to execute a helical motion (Larmor motion) around the magnetic field lines [8].

Key Confinement Parameters: Plasma Beta

To evaluate the efficiency and economic viability of a magnetic confinement system, the *plasma beta* (β) is of paramount importance. The plasma beta is defined as the ratio of the plasma thermal pressure to the magnetic pressure.

- **Low- β Regime:** In conventional toroidal devices like Tokamaks, β is typically limited to a few percent due to MHD instabilities.
- **High- β Regime:** In dipole magnetic field configurations (such as the RT-1 device), the plasma is stabilized by the strong compressibility of the magnetic flux tubes, allowing for stable confinement even when $\beta \sim 1$ [2, ?]. In this high- β regime, the plasma’s diamagnetic current significantly modifies the background magnetic field, leading to a self-organized state where the plasma pressure and magnetic field are strongly coupled [3, 6, 4, 7].

In this thesis, we focus particularly on the kinetic behavior and stability that underpin these macroscopic parameters.

2.5 RT-1 Device

2.5.1 Overview of the RT-1 (Ring Trap-1) Device

The Ring Trap-1 (RT-1), located at the University of Tokyo, is a levitated internal conductor device designed to confine plasma using a dipole magnetic field generated by a high-temperature superconducting ring. The RT-1 project aims to reproduce magnetospheric plasma confinement—similar to that observed in planetary magnetospheres like Jupiter or Earth—within a laboratory scale [2, 4].

Figure 2.2: Schematic view of the RT-1 device [2].

2.5.2 Plasma Confinement in a Dipole Magnetic Field

The most distinctive feature of RT-1 is its dipole magnetic field, generated by a superconducting coil levitated in the center of the vacuum chamber. This magnetic structure differs fundamentally from toroidal systems such as tokamaks or stellarators,

as it consists of closed dipole-shaped magnetic field lines without a toroidal field component [3].

In this dipole field, plasma particles are confined through a combination of motion along magnetic field lines, the magnetic mirror effect, and adiabatic drift motions. Because particle transport is strongly influenced by the magnetic field gradient and curvature drifts, the plasma state is governed by dominant kinetic effects, necessitating a description that accounts for the velocity space structure [5, 11].

2.5.3 Generation of High- β Plasmas

RT-1 has successfully achieved exceptionally high-beta states, reaching local $\beta \sim 1$, a value far exceeding those typically found in conventional magnetic confinement devices [6, 3]. Here, β is a dimensionless parameter representing the ratio of plasma pressure to magnetic pressure, serving as a key indicator of the interaction between the plasma and the magnetic field structure.

Under these high- β conditions, the plasma pressure significantly modifies the equilibrium magnetic field, leading to prominent nonlinear phenomena such as self-organization, turbulence transport, and the spontaneous formation of flows [3, 7]. The high-speed toroidal flows and strong velocity shears observed in RT-1 are thought to be intrinsically linked to this high- β regime and play a critical role in the stability and confinement of the system.

2.5.4 Observed Fluctuations and Unsolved Physical Phenomena

Experiments in RT-1 have revealed various low-frequency fluctuations in density and potential, as well as transitions between different electromagnetic modes [4]. While these fluctuations may be related to drift-wave turbulence driven by density gradients and velocity shear or to magnetohydrodynamic (MHD) instabilities, their precise onset mechanisms and contributions to global transport remain largely unexplained. In a system with a highly non-uniform and strong magnetic field gradient like a dipole field, the drift motion of particles and the anisotropy of the velocity distribution become essential [5]. Consequently, fluid models often prove insufficient for a complete description, highlighting the need for kinetic simulations that can capture the complex interplay between the experimental magnetic structure and the particle distribution function [15, 21].

2.6 Classical Numerical Methods for Kinetic Equations

To contextualize the advantages of quantum simulation algorithms, it is essential to understand the standard classical approaches used for solving kinetic plasma equations and their inherent limitations. The Vlasov equation represents the time evolution of the phase-space distribution function $f(\mathbf{x}, \mathbf{v}, t)$ under the influence of electromagnetic fields [8, 9, 11].

2.6.1 Method of Lines and Runge-Kutta Integration

A common approach to solving the Vlasov equation numerically is the Method of Lines (MOL). In this framework, the spatial and velocity derivatives are discretized using finite difference or finite volume methods, converting the partial differential equation (PDE) into a system of ordinary differential equations (ODEs):

$$\frac{d\mathbf{f}}{dt} = \mathcal{L}(\mathbf{f}, t), \quad (2.6.1)$$

where $\mathbf{f}$ is the discretized distribution function vector and $\mathcal{L}$ represents the semi-discretized Liouvillian operator.

For time integration, explicit schemes such as the fourth-order Runge-Kutta (RK4) method are widely employed due to their high-order accuracy and ease of implementation [10]. The RK4 scheme advances the solution from time t_n to $t_{n+1} = t_n + \Delta t$ as follows:

$$\begin{aligned} \mathbf{k}_1 &= \mathcal{L}(\mathbf{f}_n, t_n), \\ \mathbf{k}_2 &= \mathcal{L}\left(\mathbf{f}_n + \frac{\Delta t}{2}\mathbf{k}_1, t_n + \frac{\Delta t}{2}\right), \\ \mathbf{k}_3 &= \mathcal{L}\left(\mathbf{f}_n + \frac{\Delta t}{2}\mathbf{k}_2, t_n + \frac{\Delta t}{2}\right), \\ \mathbf{k}_4 &= \mathcal{L}(\mathbf{f}_n + \Delta t\mathbf{k}_3, t_n + \Delta t), \\ \mathbf{f}_{n+1} &= \mathbf{f}_n + \frac{\Delta t}{6}(\mathbf{k}_1 + 2\mathbf{k}_2 + 2\mathbf{k}_3 + \mathbf{k}_4). \end{aligned} \quad (2.6.2)$$

2.6.2 Numerical Heating and Dissipation

While explicit methods like RK4 are effective for short-term dynamics, they suffer from significant drawbacks in long-term plasma simulations, particularly for studying self-organization phenomena or low-frequency fluctuations such as those observed in the RT-1 device [3, 4].

The primary issue is that standard explicit integrators are generally non-symplectic and non-unitary. Consequently, they do not strictly preserve the geometric invariants

of the Hamiltonian system, such as the total energy and the phase-space volume (L^2 -norm of f) [12]. This leads to two critical artifacts:

- **Numerical Dissipation:** The amplitude of the distribution function may decay over time, mimicking artificial collisions.
- **Non Conservation of Energy:** The total energy of the system often drifts monotonically [11].

Although symplectic integrators exist for particle-based simulations (e.g., the Boris algorithm for particle orbits [12]), developing strictly conservative schemes for the high-dimensional grid-based Vlasov equation remains computationally expensive. In contrast, the quantum algorithms proposed in this thesis rely on unitary evolution operators e^{-iHt} , which inherently conserve the L^2 -norm and suppress numerical heating, offering a robust framework for long-time integration [15, 21, 19].

Chapter 3

Problem Definition

3.1 Background and Previous Research

3.1.1 Kinetic Effects and Simulation Challenges in Magnetically Confined Plasmas

In magnetically confined plasmas, fluid approximations often prove insufficient for analyzing phenomena where velocity space anisotropy or resonant particle effects play a dominant role, such as high- β turbulence and wave-particle interactions. In these regimes, a kinetic description based on the Vlasov–Maxwell system provides the necessary physical fidelity [8, 9]. However, numerical simulations of the Vlasov equation present a formidable computational challenge due to the exponential scaling of computational complexity with respect to phase-space degrees of freedom. Since the distribution function $f(\boldsymbol{x}, \boldsymbol{v}, t)$ evolves in a six-dimensional phase space, grid-based methods require memory and processing power that scale as N^6 (where N is the number of grid points per dimension), making high-resolution simulations computationally intractable on classical hardware.

Conventional approaches, such as grid-based Eulerian methods or Particle-in-Cell (PIC) techniques, have been widely utilized [11]. However, standard explicit time-integration schemes used in these methods (e.g., Runge-Kutta methods) are generally non-unitary. In long-term simulations essential for studying transport and self-organization, these non-unitary schemes inevitably introduce numerical dissipation or energy loss (secular energy drift) [12]. This accumulation of numerical errors obscures delicate physical mechanisms, creating a critical need for a computational methodology that guarantees rigorous numerical stability over extended integration periods.

3.1.2 High- β Physics and Long-Term Confinement in the RT-1 Device

The Ring Trap 1 (RT-1) device, employing a levitated superconducting dipole magnet, offers a unique platform for investigating high- β plasmas relevant to planetary magnetospheres [2]. Experiments in RT-1 have achieved local β values exceeding 70% [6] and have demonstrated the self-organization of peaked density profiles driven by inward turbulent diffusion [3, 5, 7].

A distinctive feature of RT-1 is its long particle confinement time, which significantly exceeds the characteristic kinetic timescales (e.g., cyclotron or bounce periods). Observing the relaxation processes and structural formation in such systems requires stable time integration over a vast number of time steps. Recently, transitions from electrostatic to electromagnetic fluctuation modes have been observed, suggesting complex couplings between particle kinetics and electromagnetic fields [4]. Unlike low- β tokamaks where the magnetic field is rigid, the magnetic field in RT-1 is significantly modified by plasma pressure. To elucidate the kinetic mechanisms underlying these transport phenomena and stability limits, advanced simulation tools are required that can handle the specific magnetic geometry and maintain numerical accuracy over the long confinement timescales characteristic of the device.

3.1.3 Quantum Computing for Kinetic Plasma Simulation

Quantum computing offers a promising paradigm for simulating high-dimensional physical systems [13, 14]. Quantum Hamiltonian simulation algorithms allow for the time evolution of a system with a cost that scales polylogarithmically with the dimension of the state space, providing a potential solution to the intractable scaling issues inherent in classical Vlasov simulations [15].

While early quantum algorithms focused on sparse Hamiltonians [19], discretized Vlasov operators in realistic geometries often possess complex, non-sparse structures. Recent breakthroughs in Quantum Signal Processing (QSP) [16] and the Quantum Singular Value Transformation (QSVT) [17, 18] have unified quantum algorithms, enabling the efficient simulation of general matrices via block encoding [20].

A fundamental challenge in applying these techniques to plasma physics is the inherent nonlinearity of the Vlasov–Maxwell system. While the quantum processor evolves states linearly, plasma dynamics involve a self-consistent coupling between the distribution function and electromagnetic fields. Building on the linear instability analysis by Higuchi et al. [21], this study adopts a quantum–classical hybrid approach. In this framework, the computationally intensive time evolution of the distribution function is handled by the quantum processor using unitary operators, while the electromagnetic fields are updated classically. This methodology leverages the stability of quantum unitary evolution—essential for the long confinement times of RT-1—while accommodating the nonlinear coupling inherent in high- β plasmas.

3.2 Research Objectives

The primary objective of this research is to construct and verify a numerically stable hybrid quantum-classical simulation framework for the Vlasov–Poisson system, modeling the plasma dynamics in a magnetic mirror configuration relevant to the Ring Trap 1 (RT-1) device. By isolating the algorithmic properties of unitary evolution from hardware noise, this study focuses on the following specific goals:

1. **Implementation of Unitary Time Evolution for the 1D1V Vlasov Equation**

We aim to apply the Quantum Singular Value Transformation (QSVT) [17] to the time evolution of the Vlasov equation in a 1D1V phase space (one spatial and one velocity dimension). Unlike classical finite difference methods where discretization errors lead to non-physical numerical diffusion [11], the quantum approach evolves the system via unitary operators [19, 20]. A key goal is to demonstrate that this unitarity strictly preserves number of particles and suppressing energy loss even in long-term simulations [15].

2. **Development of a Hybrid Vlasov–Poisson Framework**

To address the nonlinear coupling between the plasma distribution and the self-consistent electric field, we propose a quantum-classical hybrid framework. We construct an iterative solver where the phase space evolution is handled by the quantum algorithm (Vlasov solver), while the electric field is updated classically. Specifically, we employ a spectral Poisson solver based on the Fast Fourier Transform (FFT) to compute the electrostatic potential from the density profile at each time step. This approach reconciles the linearity of quantum mechanics with the nonlinear nature of collective plasma effects.

3. **Modeling Magnetic Mirror Confinement in RT-1**

We apply the proposed method to a model simulating the dipole magnetic field structure of the RT-1 device. Instead of solving the full Maxwell equations, the magnetic field is treated as a fixed background mirror field $B(z)$, effectively incorporated into the Hamiltonian as a magnetic mirror force $F = -\mu\nabla B$. By initializing the system with conditions representing trapped particles in this mirror potential [6], we investigate whether the quantum solver can accurately reproduce confinement dynamics without the secular energy drift observed in classical schemes [3].

4. **Validation via Numerical Emulation of Unitary Operators**

Instead of executing probabilistic simulations on a noisy quantum circuit simulator, we perform numerical experiments by calculating the matrix-vector multiplication equivalent to the quantum unitary operators. While this approach idealizes the polynomial approximation errors inherent in Quantum Signal Processing (QSP), it provides a rigorous mathematical testbed to verify the conser-

vation laws (unitarity) of the proposed algorithm [1]. By comparing the time evolution of the distribution function vector against standard classical solvers (e.g., RK4), we aim to quantitatively demonstrate the algorithmic advantage of unitary evolution in suppressing numerical dissipation and energy loss.

Through these objectives, this thesis establishes a foundational methodology for applying quantum computing to kinetic plasma problems, offering a robust alternative to classical methods for future high-fidelity simulations of fusion plasmas.

3.3 Limitations of This Study

This study successfully demonstrated the numerical stability and conservation properties of a quantum–classical hybrid framework for the Vlasov–Poisson system in a magnetic mirror configuration. However, several critical constraints and limitations remain to be addressed for the realization of practical fusion plasma simulations.

3.3.1 Constraints on Computational Scale and Dimensions

The numerical experiments in this research were restricted to a 1D1V (one spatial and one velocity dimension) model. This reduction focuses on the fundamental bounce motion of trapped particles along the magnetic field lines of the RT-1 device. However, it inherently neglects cross-field drifts, finite Larmor radius effects, and toroidal dynamics specific to the actual 3D dipole geometry [2]. Furthermore, the problem size (number of grid points) was limited by the computational resources required for the classical emulation of quantum operators. Since the classical simulation of the distribution function vector involves dense matrix-vector multiplications or full state-vector storage, the memory scaling restricts the resolution. Extending this approach to a realistic 2D3V or 3D3V phase space requires actual fault-tolerant quantum hardware to leverage the logarithmic scaling advantage [1].

3.3.2 Idealized Numerical Emulation and Hardware Noise

In this study, the unitary time evolution operators were verified via classical matrix-vector multiplication rather than execution on a quantum circuit simulator or real hardware. This approach isolates the algorithmic properties of the method, effectively treating the system as an ideal, noise-free quantum computer. Consequently, the effects of physical noise, gate errors, decoherence, and limited qubit connectivity—which are dominant in the Noisy Intermediate-Scale Quantum (NISQ) era—were excluded from this analysis. While this confirms the theoretical validity of the algorithm, a quantitative evaluation of the implementation overhead, such as the circuit depth required for block encoding and the number of ancilla qubits, remains a subject for future work [18].

3.3.3 Approximation Errors in Quantum Signal Processing

The implementation of the time evolution operator e^{-iHt} relies on polynomial approximations (e.g., Jacobi–Anger expansion) within the framework of Quantum Signal Processing (QSP). Although the matrix emulation assumes high-precision arithmetic, in a real quantum implementation, finite-degree truncation inevitably introduces approximation errors. The trade-off between the “accuracy of the unitary operator” (determined by the polynomial degree) and the “circuit complexity” needs to be systematically optimized, especially when extending the simulation to longer timescales where phase errors might accumulate [16, 19].

3.3.4 Global Unitarity in the Hybrid Framework

A fundamental limitation of the proposed hybrid framework lies in the treatment of nonlinearity. While the quantum Vlasov solver guarantees strict unitarity (norm conservation) for the distribution function update, the electric field is calculated classically using a spectral Poisson solver at each time step. This iterative coupling between the quantum state and the classical field breaks the global symplectic structure of the system [15]. Consequently, although energy loss is significantly suppressed compared to fully classical explicit schemes, strict energy conservation is not theoretically guaranteed for the coupled Vlasov–Poisson system. The long-term stability of this semi-classical coupling relies on the time-step size Δt and requires further rigorous mathematical analysis [21].

3.3.5 Simplification of Physical Processes in RT-1

To prioritize algorithmic verification, the physical model of the RT-1 plasmas was simplified to a collisionless electrostatic model in a fixed magnetic mirror field. We treated the magnetic field as a static background force potential, omitting the self-consistent modification of the magnetic field by the high- β plasma pressure [6]. Moreover, complex electromagnetic fluctuations observed in recent experiments, such as the transition to electromagnetic modes [4], involve multi-scale interactions that are not captured in the current 1D1V electrostatic formulation.

3.3.6 Verification of Computational Advantage

This study focused on numerical stability and algorithm construction, rather than demonstrating an actual computational speedup. Since the verification was performed via classical matrix emulation, the execution time scales polynomially with the system size, which is slower than efficient classical algorithms. While theoretical analysis suggests an exponential advantage in memory and a polylogarithmic speedup in time with respect to the phase-space dimension [14], verifying this scaling advantage requires implementation on actual quantum hardware with sufficient logical qubits.

Chapter 4

Methodology

4.1 Hybrid Simulation Framework for the 1D1V Vlasov–Poisson System

To investigate the numerical stability and conservation properties of kinetic plasma simulations, we employ a quantum-classical hybrid scheme applied to a 1D1V (one spatial and one velocity dimension) Vlasov–Poisson system. In this study, our primary focus is not on demonstrating computational speedup (quantum advantage), but rather on verifying the mathematical structure of the quantum algorithm. Specifically, we aim to demonstrate that the strictly unitary time evolution provided by quantum algorithms can suppress the numerical diffusion and energy loss inherent in classical explicit difference schemes.

The simulation loop proceeds in discrete time steps Δt . Let $f^{(n)}(z, v)$ denote the distribution function along the magnetic field line z , and $E^{(n)}(z)$ denote the self-consistent electric field at time step n . The overall algorithm is illustrated as follows:

1. **Initialization:** Prepare the initial state vector $|\psi^{(0)}\rangle$ corresponding to the distribution function $f_0(z, v)$. To model the high- β plasma in RT-1, the spatial profile is initialized with a density peak at the magnetic equator ($z = 0$), confined by the magnetic mirror potential, combined with a Maxwellian velocity distribution.
2. **Hamiltonian Construction (Classical):** At each time step n , construct the discretized Vlasov Hamiltonian matrix $H_{Vlasov}^{(n)}$ on the classical computer. The Hamiltonian governing the 1D1V dynamics is given by:

$$H_{Vlasov}^{(n)} = v \frac{\partial}{\partial z} + \left(\frac{q}{m} E^{(n)}(z) + \frac{F_{mirror}(z)}{m} \right) \frac{\partial}{\partial v}, \quad (4.1.1)$$

where $E^{(n)}(z)$ is the electric field updated iteratively, and $F_{mirror}(z) = -\mu \nabla B(z)$ is the static magnetic mirror force derived from the fixed background dipole field $B(z)$.

3. **Block-Encoding (Classical $\rightarrow$ Quantum):** Embed the non-unitary Hamiltonian matrix $H_{Vlasov}^{(n)}$ into a unitary operator U_H using block-encoding techniques. This step mathematically maps the dissipative classical operator into a larger unitary framework, preparing the oracle required for the quantum algorithm.
4. **Time Evolution via QSVT (Quantum):** Apply the time evolution operator $U_{ev} \approx e^{-iH_{Vlasov}^{(n)}\Delta t}$ to the quantum state:

$$|\psi^{(n+1)}\rangle = U_{ev} |\psi^{(n)}\rangle. \quad (4.1.2)$$

The operator $e^{-iH\Delta t}$ is approximated by a polynomial of H using the Quantum Singular Value Transformation (QSVT). Crucially, this operation is strictly unitary, ensuring that the L^2 -norm of the state is conserved, which theoretically prevents particle reduction.

5. **Density Extraction (Quantum $\rightarrow$ Classical):** From the updated state $|\psi^{(n+1)}\rangle$, calculate the spatial charge density distribution $\rho^{(n+1)}(z) = q \int f^{(n+1)} dv$. While a real quantum implementation would require Quantum Amplitude Estimation (QAE) for readout, this study focuses on algorithmic verification; thus, we compute the density directly from the state vector via matrix operations.
6. **Poisson Solver (Classical):** Using the extracted charge density $\rho^{(n+1)}$, update the electric field to $E^{(n+1)}$ by solving Poisson's equation on the classical grid. We employ a spectral solver based on the Fast Fourier Transform (FFT) to strictly enforce periodic boundary conditions and ensure spatial accuracy:

$$E^{(n+1)}(z) = \mathcal{F}^{-1} \left[\frac{\mathcal{F}[\rho^{(n+1)}]}{ik} \right]. \quad (4.1.3)$$

7. **Loop:** Repeat steps 2–6 to strictly simulate the nonlinear interaction between the phase-space fluid and the electrostatic field until the target simulation time is reached.

4.1.1 Verification via Numerical Emulation

The detailed quantum circuit architecture designed to implement the block-encoding and QSVT time evolution is presented in Section 2. This schematic represents the sequence of quantum gates required to execute the algorithm on a fault-tolerant quantum computer.

However, it is important to clarify that this study performs a *numerical emulation* based on linear algebra, rather than executing the circuits on a simulator or hardware. The unitary operator U_{ev} , corresponding to the total action of the quantum circuit,

was explicitly constructed as a dense matrix on a classical computer, and the time evolution was computed via:

$$\psi^{(n+1)} = \mathbf{U}_{ev} \cdot \psi^{(n)}. \quad (4.1.4)$$

Since our objective is to verify whether the *mathematical structure* of the unitary algorithm can suppress numerical diffusion and heating, this classical emulation is sufficient and appropriate. It allows for the rigorous evaluation of conservation properties in an ideal environment, isolating the algorithmic performance from hardware-specific noise or speed limitations.

4.2 Initialization and Hamiltonian Construction

In this section, we describe the specific numerical procedures for preparing the initial quantum state and constructing the Hamiltonian matrix for the 1D1V Vlasov–Poisson system under a magnetic mirror configuration. These procedures correspond directly to the implemented numerical code.

4.2.1 Initial State Preparation

The simulation domain is discretized into N_z spatial grid points and N_v velocity grid points. The quantum state vector $|\psi\rangle$ represents the square root of the phase-space probability distribution function.

We initialize the system with a distribution that mimics a high- β plasma core confined in the RT-1 device. The initial distribution function $f_0(z, v)$ is constructed as the product of a spatial Gaussian profile located at the magnetic equator ($z = 0$) and a velocity-space Maxwellian distribution:

$$f_0(z, v) = n(z) \cdot f_M(v) = \exp\left(-\frac{z^2}{2\sigma_z^2}\right) \cdot \exp\left(-\frac{v^2}{2v_{th}^2}\right), \quad (4.2.1)$$

where σ_z represents the spatial spread of the plasma clump, and v_{th} is the thermal velocity.

The quantum state amplitude at grid index (i_z, i_v) is initialized as the square root of this distribution:

$$\psi_{i_z, i_v} = \sqrt{n(z_{i_z}) \cdot f_M(v_{i_v})}. \quad (4.2.2)$$

Finally, the state vector is normalized such that its L^2 -norm equals unity ($\|\psi\| = 1$), satisfying the probability interpretation of quantum mechanics.

4.2.2 Magnetic Mirror Field and Effective Force

To model the magnetic mirror confinement characteristic of a dipole field, we define a background magnetic field $B(z)$ that is minimal at the center and increases towards

the boundaries. We adopt a parabolic profile given by:

$$B(z) = B_0(1 + \alpha z^2), \quad (4.2.3)$$

where B_0 is the central magnetic field strength and α determines the steepness of the mirror confinement.

In the 1D1V guiding-center approximation, particles experience a mirror force F_{mirror} proportional to the gradient of the magnetic field. Assuming a constant magnetic moment μ , the force is calculated as:

$$F_{mirror}(z) = -\mu \frac{dB}{dz} = -2\mu B_0 \alpha z. \quad (4.2.4)$$

In the numerical implementation, this gradient is computed using finite differences or analytical derivatives of the B -field array.

4.2.3 Construction of the Discretized Hamiltonian

The time evolution of the Vlasov system is governed by a Hamiltonian matrix H constructed from the advection and force terms. The total force acting on the particles at grid point z_i is the sum of the static mirror force and the self-consistent electric force:

$$F_{total}(z_i) = qE(z_i) + F_{mirror}(z_i). \quad (4.2.5)$$

The Hamiltonian matrix elements are determined using a second-order central difference scheme to ensure the operator is Hermitian (or anti-Hermitian depending on the convention, here mapped to a Hermitian quantum operator). Let $\text{idx}(i, j)$ denote the flattened index for the grid point (z_i, v_j) . The non-zero elements of the Hamiltonian H corresponding to the Vlasov equation $\partial_t f + v \partial_z f + F \partial_v f = 0$ are constructed as follows:

1. Spatial Advection Term ($v \partial_z$): For each velocity v_j , the coupling between adjacent spatial grid points is given by:

$$H_{\text{idx}(i,j), \text{idx}(i+1,j)} = +i \frac{v_j}{2\Delta z}, \quad (4.2.6)$$

$$H_{\text{idx}(i,j), \text{idx}(i-1,j)} = -i \frac{v_j}{2\Delta z}. \quad (4.2.7)$$

2. Velocity Acceleration Term ($F \partial_v$): For each spatial position z_i , using the local total force $F_{local} = F_{total}(z_i)$, the coupling between adjacent velocity grid points is:

$$H_{\text{idx}(i,j), \text{idx}(i,j+1)} = +i \frac{F_{local}}{2\Delta v}, \quad (4.2.8)$$

$$H_{\text{idx}(i,j), \text{idx}(i,j-1)} = -i \frac{F_{local}}{2\Delta v}. \quad (4.2.9)$$

Here, i represents the imaginary unit. This construction ensures that H is a Hermitian matrix with purely imaginary off-diagonal elements (and zeros on the diagonal), which corresponds to a unitary time evolution operator $U = e^{-iH\Delta t}$ suitable for quantum simulation.

4.3 Hamiltonian Block-Encoding

A critical step in our method is the mapping of the classical discretized Hamiltonian matrix into a quantum circuit. As derived in Section 4.2.3, the discretized Vlasov equation can be written as:

$$\frac{\partial f}{\partial t} = -iH_{Vlasov}f, \quad (4.3.1)$$

where H_{Vlasov} is a sparse Hermitian matrix representing the advection in phase space. We utilize the Block-Encoding formalism. A unitary U is a (α, a, ϵ) -block-encoding of H if:

$$\|H - \alpha(\langle 0|^{\otimes a} \otimes I)U(|0\rangle^{\otimes a} \otimes I)\| \leq \epsilon. \quad (4.3.2)$$

Here, α is the sub-normalization factor (related to the spectral norm of H) and a is the number of ancilla qubits.

The Vlasov Hamiltonian consists of spatial advection terms ($v_x \partial_x$, etc.) and force advection terms ($F_x \partial_{v_x}$, etc.). Since we use a central difference scheme, these derivative operators can be decomposed into sums of shift operators. The Hamiltonian takes the form:

$$H_{Vlasov} = \sum_k c_k P_k, \quad (4.3.3)$$

where P_k are Pauli strings or permutation matrices implementing the shifts, and c_k are coefficients determined by the velocity grid and the electromagnetic fields. We implement this using the PREP and SEL circuits standard in LCU literature. The PREP circuit encodes the coefficients c_k into the amplitudes of ancilla qubits, while the SEL circuit applies the corresponding shift operations controlled by the ancilla.

4.4 Quantum Singular Value Transformation (QSVT)

To implement the time evolution $e^{-iH\Delta t}$, we use QSVT. This method implements a polynomial transformation $P(x) \approx e^{-ix\Delta t}$ to the singular values of the block-encoded Hamiltonian.

We employ the Jacobi-Anger expansion to approximate the exponential function:

$$e^{-ix\tau} \approx J_0(\tau) + 2 \sum_{k=1}^d (-i)^k J_k(\tau) T_k(x), \quad (4.4.1)$$

where J_k are Bessel functions of the first kind and T_k are Chebyshev polynomials. For a simulation time step Δt and desired precision ϵ , the degree of the polynomial d scales as $\mathcal{O}(\tau + \log(1/\epsilon))$, where $\tau = \|\alpha H\| \Delta t$.

The QSVT circuit is constructed by interleaving the block-encoding unitary U_H and its inverse $U_H^\dagger$ with single-qubit signal processing rotations R_ϕ . The phase angles $\phi = \{\phi_0, \dots, \phi_d\}$ are pre-computed classically to synthesize the desired polynomial $P(x)$.

4.5 Measurement and Classical Feedback

The bottleneck of hybrid simulations is often the data transfer from quantum to classical processors. To update the self-consistent field, we require the macroscopic charge density profile $\rho(z)$. In the 1D1V phase space discretized into $N_z \times N_v$ grid points, the quantum state vector ψ encodes the probability amplitudes such that $|\psi_{i,j}|^2$ corresponds to the distribution function $f(z_i, v_j)$.

The charge density at spatial grid point z_i is obtained by summing the probability distribution over the velocity space:

$$\rho(z_i) = q \sum_{j=0}^{N_v-1} |\psi_{\text{idx}(i,j)}|^2 \Delta v. \quad (4.5.1)$$

In a full quantum implementation, this summation corresponds to measuring the expectation value of a local density operator. To achieve the precision δ required for stable feedback, Quantum Amplitude Estimation (QAE) provides a quadratic speedup ($\mathcal{O}(1/\delta)$) compared to classical sampling ($\mathcal{O}(1/\delta^2)$). In our numerical emulation, we compute this sum directly from the state vector.

The measured density $\rho(z)$ is then fed into the classical Poisson solver. To ensure numerical stability and strictly enforce periodic boundary conditions, we employ a spectral method based on the Fast Fourier Transform (FFT). The procedure is as follows:

1. **Neutralization:** To model a quasi-neutral plasma and ensure solvability of Poisson's equation under periodic boundaries (where the total charge must be zero), we subtract the spatial mean of the ion background charge density:

$$\delta\rho(z) = \rho(z) - \langle \rho \rangle_z. \quad (4.5.2)$$

2. **Spectral Solution:** We transform the density to Fourier space, $\tilde{\rho}_k = \mathcal{F}[\delta\rho(z)]$. The electrostatic potential ϕ satisfies $-k^2 \tilde{\phi}_k = \tilde{\rho}_k / \epsilon_0$. The electric field $E = -\nabla\phi$ is then computed directly in k-space:

$$\tilde{E}_k = \frac{\tilde{\rho}_k}{ik\epsilon_0} \quad (k \neq 0), \quad \tilde{E}_{k=0} = 0. \quad (4.5.3)$$

3. **Inverse Transform:** The physical electric field is obtained via the inverse Fourier transform:

$$E(z) = g \cdot \text{Re} \left(\mathcal{F}^{-1}[\tilde{E}_k] \right). \quad (4.5.4)$$

Here, g is a coupling constant (set to unity in standard physical units) that allows for tuning the strength of the electrostatic interaction (plasma frequency) relative to the kinetic energy in numerical experiments.

This updated electric field $E(z)$ is then used to construct the Hamiltonian for the subsequent time step, completing the self-consistent feedback loop.

4.6 Simulation Setup for the RT-1 device

To evaluate the validity of the proposed hybrid framework, we configure the simulation parameters to model the fundamental particle confinement physics in the Ring Trap 1 (RT-1) device. Specifically, we reduce the problem to a 1D1V system describing the bounce motion of charged particles along a magnetic field line. The specific setup is as follows:

- **Magnetic Field Configuration:** Instead of the full 3D dipole geometry, we employ a 1D magnetic mirror model that approximates the field strength variation along a field line near the magnetic equator. The background magnetic field $B(z)$ is defined as a parabolic function:

$$B(z) = B_0(1 + \alpha z^2), \quad (4.6.1)$$

where z represents the coordinate along the field line, B_0 is the field strength at the equator ($z = 0$), and α characterizes the steepness of the mirror field [2]. This generates a restoring magnetic mirror force $F_{\text{mirror}} \propto -z$, creating a "magnetic bottle" effect.

- **Phase Space Discretization and Quantum Encoding:** The phase space is discretized into $N_z \times N_v$ grid points. The quantum state vector $|\psi\rangle$ is defined in the computational basis $|i\rangle_z |j\rangle_v$, where indices i and j correspond to spatial position z_i and velocity v_j , respectively. Crucially, the complex amplitude $c_{i,j}$ of the quantum state $|\psi\rangle = \sum_{i,j} c_{i,j} |i\rangle_z |j\rangle_v$ encodes the square root of the normalized distribution function:

$$c_{i,j} = \sqrt{f(z_i, v_j) \Delta z \Delta v}. \quad (4.6.2)$$

Consequently, the probability of measuring the basis state $|i\rangle_z |j\rangle_v$, given by $|c_{i,j}|^2$, corresponds directly to the phase-space probability density of the plasma.

- **Initial Condition:** Reflecting the high- β operations in RT-1 where plasma is strongly confined in the core region [6], the initial distribution is modeled as a product of a spatial Gaussian profile and a velocity Maxwellian:

$$f_0(z, v) \propto \exp\left(-\frac{z^2}{2\sigma_z^2}\right) \exp\left(-\frac{v^2}{2v_{th}^2}\right). \quad (4.6.3)$$

Here, σ_z represents the spatial extent of the plasma clump, and v_{th} denotes the thermal velocity. This setup allows us to verify whether the quantum unitary evolution can stably maintain this equilibrium state without the artificial diffusion typically observed in classical Vlasov solvers [11].

Chapter 5

Numerical Results

In this chapter, we present the numerical simulation results of the 1D1V (one-dimensional in space and one-dimensional in velocity space) Vlasov–Maxwell system, obtained using the proposed hybrid quantum-classical algorithm. The primary objective of this numerical experiment is to demonstrate the superior precision of the Quantum Singular Value Transformation (QSVT) compared to classical methods, specifically by highlighting its ability to perform time evolution free from numerical diffusion and energy loss.

5.1 Simulation Settings

5.1.1 Grid Configuration and Quantum Resource Allocation

For the simulation, we set the spatial grid to $N_x = 32$ and the velocity grid to $N_v = 32$. Consequently, the total number of grid points in the phase space is $N = 32 \times 32 = 1024$. From a quantum computing perspective, the number of qubits required to encode this state is $n = \log_2(1024) = 10$ qubits. Given the current capabilities of quantum simulators, this scale is both realistic and challenging, as it allows for the simultaneous validation of Hamiltonian simulation and self-consistent field calculations. However, due to the significant circuit depth inherent to QSVT, classical simulation becomes prohibitively time-consuming. Therefore, we do not employ a quantum simulator in this study.

5.1.2 Initial Conditions and Physical Parameters

The initial distribution function $f(x, v, t = 0)$ is defined as follows:

$$f(x, v, 0) = n_0 \exp\left(-\frac{x^2}{2\sigma_x^2}\right) \cdot \exp\left(-\frac{v^2}{2v_{th}^2}\right) \quad (5.1.1)$$

where the spatial distribution follows a Gaussian profile peaked near the magnetic equator ($\sigma_x = 0.2$), and the velocity space follows a Maxwellian distribution with a

thermal velocity $v_{th} = 1.0$. An inhomogeneous background magnetic field modeling a magnetic mirror is applied, defined by the parabolic profile $B(z) = B_0(1 + \alpha z^2)$ with $B_0 = 1.0$ and $\alpha = 0.5$. And the initial value of the self-induced electric field E_x is set to zero.

5.2 Numerical Results and Analysis

In this section, we present the simulation results of the 1D1V Vlasov–Poisson system under a magnetic mirror configuration. We compare the proposed hybrid quantum algorithm against a classical fourth-order Runge-Kutta (RK4) integrator to evaluate physical fidelity, numerical stability, and conservation properties.

5.2.1 Phase Space Dynamics and Physical Validation

Figure 5.1 illustrates the time evolution of the distribution function $f(z, v_z)$ in phase space at $t = 0.0$, $t = 0.5$, and $t = 1.0$. The system is initialized with a Gaussian-Maxwellian distribution centered at the magnetic equator ($z = 0$). As time evolves, the particles undergo bounce motion confined by the magnetic mirror force $F_{mirror} \propto -z$. The results show the characteristic *phase space shearing*, where the distribution tilts and stretches into a filamentary structure. This behavior confirms that the quantum algorithm and classic algorithm correctly implements the kinetic advection terms via the block-encoded Hamiltonian operators. The smooth evolution in the early phase ($t \leq 1.0$) demonstrates that the grid resolution ($N_z = 32, N_v = 32$) is sufficient to capture the macroscopic bounce dynamics.

5.2.2 Quantitative Stability Analysis

To rigorously evaluate the conservation properties, we monitored the total particle number $N(t)$ and kinetic energy per particle $\mathcal{E}(t)$ over both short, middle and long timescales.

Short-term Dynamics and Energy Discrepancy ($t \leq 1.0$)

Figure 5.2 illustrates the quantitative comparison during the initial phase ($t = 0$ to 1.0).

- **Particle Number Conservation:** The quantum algorithm (Blue Solid) maintains the normalized particle number strictly at 1.00 as a direct consequence of unitary evolution. While the classical RK4 method (Red Dashed) begins to exhibit a downward drift around $t = 0.6$, the particle loss remains relatively moderate ($\sim 0.02\%$) within this time window.

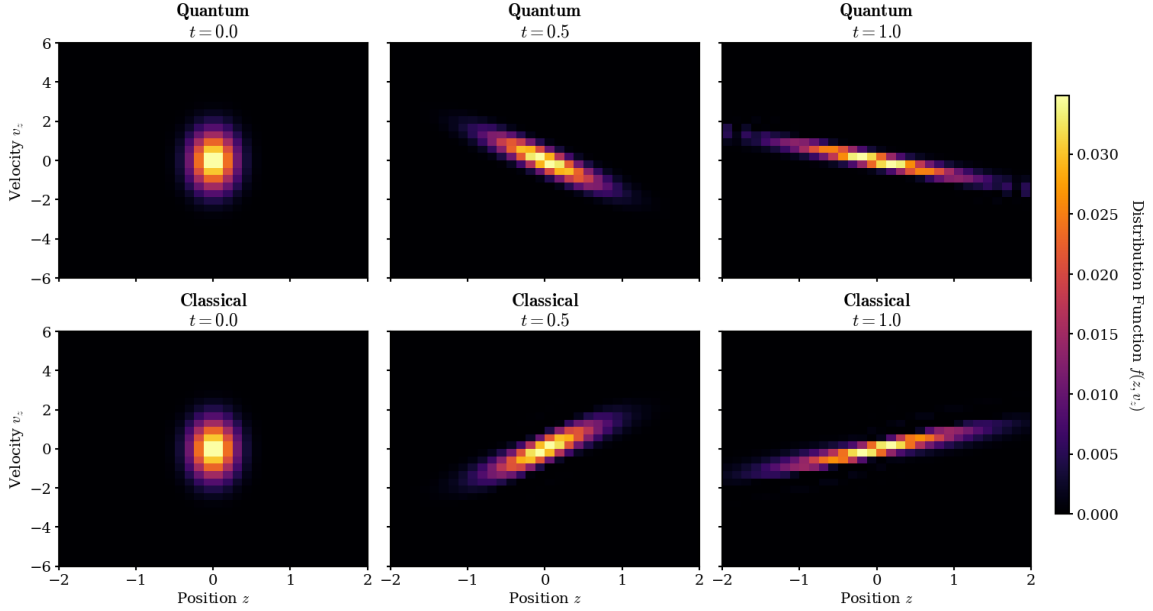


Figure 5.1: Phase space evolution of the distribution function $f(z, v_z)$ at $t = 0.0, 0.5, 1.0$. The quantum and classical results show excellent agreement in capturing the initial phase space shearing caused by the magnetic mirror field.

- **Kinetic Energy per Particle:** A more significant discrepancy is observed in the kinetic energy evolution (Fig. 5.2(b)). Even in this early regime where the particle number is reasonably preserved, the classical result deviates markedly from the quantum benchmark starting at $t = 0.4$. The classical scheme suffers from excessive energy decay (numerical cooling), resulting in a kinetic energy value of approximately 0.15 at $t = 1.0$, compared to ~ 0.21 in the quantum simulation. This indicates that non-physical dissipation in the classical integrator distorts the dynamical energy balance significantly earlier than it affects mass conservation.

Intermediate Dynamics: Magnetic Mirror Reflection ($t \leq 2.0$)

Figure 5.3 captures the dynamics of the first magnetic mirror reflection, where the plasma interacts with the high-field region.

- **Kinetic Energy and Numerical Friction:** The temporal evolution of the kinetic energy per particle (Fig. 5.3(b)) clearly illustrates the magnetic mirror effect. The initial decrease corresponds to the conversion of kinetic energy into magnetic potential energy as particles climb the magnetic gradient. The subsequent turnaround at $t \approx 1.6$ signifies the reflection (bounce) of particles back towards the center.

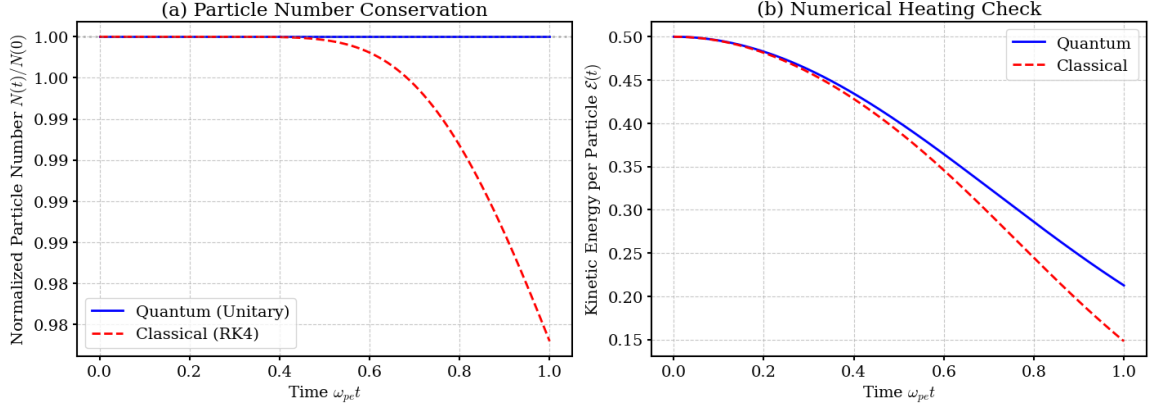


Figure 5.2: Short-term quantitative comparison ($t = 0$ to 1.0). (a) Normalized particle number $N(t)/N(0)$ and (b) Kinetic energy per particle. While the classical RK4 method (Red Dashed) exhibits a slight drift in particle number starting at $t \approx 0.6$, a more significant deviation appears earlier in the kinetic energy (at $t \approx 0.4$), where the classical solver suffers from severe numerical cooling (non-physical dissipation) compared to the unitary quantum simulation (Blue Solid).

- **Spurious Particle Fluctuations:** Regarding particle conservation (Fig. 5.3(a)), the quantum algorithm maintains exact stability. Conversely, the classical result exhibits a problematic artifact: after a significant drop ($> 4\%$) reaching a minimum at $t \approx 1.5$, it shows a spurious increase in particle number. Since the physical system is closed, this "recovery" is non-physical and indicates that the numerical errors are not merely diffusive but are introducing unpredictable fluctuations that violate the fundamental continuity equation.

Long-term Stability and Classical Breakdown ($t > 4.0$)

Figure 5.4 extends the simulation to $t \approx 4.0$ to test the limits of numerical stability. This comparison highlights the critical robustness gap between the two methods.

- **Catastrophic Breakdown due to Numerical Dissipation:** The classical solver (Red Dashed) maintains marginal stability only until $t \approx 9.2$. At this point, the cumulative effect of non-physical dissipation destabilizes the phase space structure, leading to a scenario where the simulation collapses.

It is imperative to correctly interpret the kinetic energy plot behavior: the classical curve exhibits a sharp singularity and then abruptly flatlines at zero. This zero-value phase does not represent a physical relaxation to a ground state. Rather, it is a **post-processing artifact** where the output is clamped to zero because the denominator—the total particle number $N(t)$ —has collapsed to zero or negative values. This confirms that the classical scheme suffers from a total

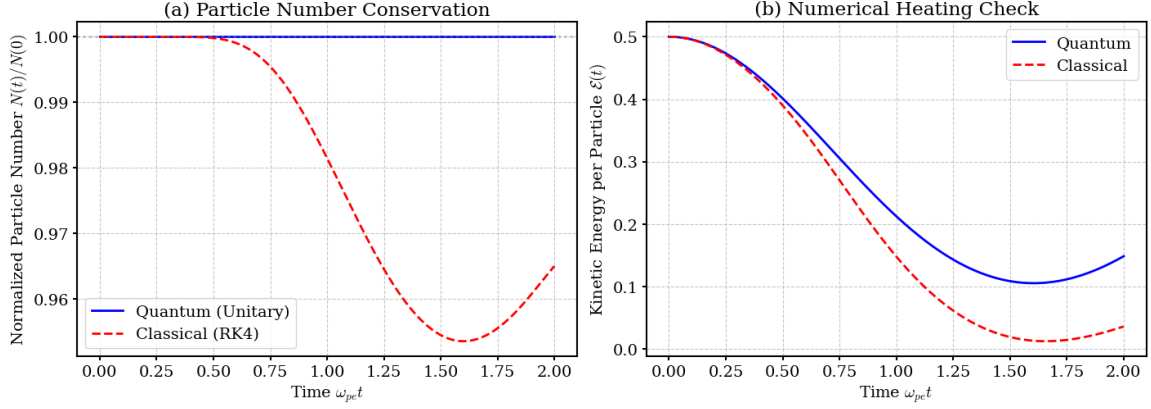


Figure 5.3: Intermediate dynamics during the first magnetic mirror reflection ($t = 0$ to 2.0). (a) Normalized particle number and (b) Kinetic energy per particle. The classical method (Red Dashed) exhibits non-physical artifacts, including spurious particle fluctuations and severe numerical friction (incomplete energy recovery) during the bounce phase. In contrast, the quantum simulation (Blue Solid) demonstrates symmetric, reversible dynamics characteristic of Hamiltonian mechanics.

breakdown of the plasma density distribution, rendering the results physically meaningless beyond this point.

- **Unconditional Stability of Quantum Method:** In stark contrast, the quantum simulation (Blue Solid) remains strictly stable throughout the entire duration. The particle number is preserved to machine precision due to the unitary nature of the time evolution operator $U = e^{-iH\Delta t}$. The kinetic energy oscillates within a bounded physical range without exhibiting the irreversible decay characteristic of the classical result. This demonstrates that the quantum algorithm inherently prevents the structural collapse caused by numerical dissipation, ensuring reliability for long-term fusion plasma analysis.

5.3 Limitations and Future Outlook

The numerical results presented in this study demonstrate the fundamental stability of the QSVT-based hybrid algorithm, particularly in contrast to the catastrophic breakdown driven by numerical dissipation in classical explicit solvers. However, several challenges remain for scaling this approach to practical fusion reactor simulations.

5.3.1 Resolution Limits and Phase Space Delocalization

While the current grid resolution of $N_z \times N_v = 32 \times 32$ (requiring 10 logical qubits) was sufficient to capture the macroscopic bounce motion and initial phase space shearing,

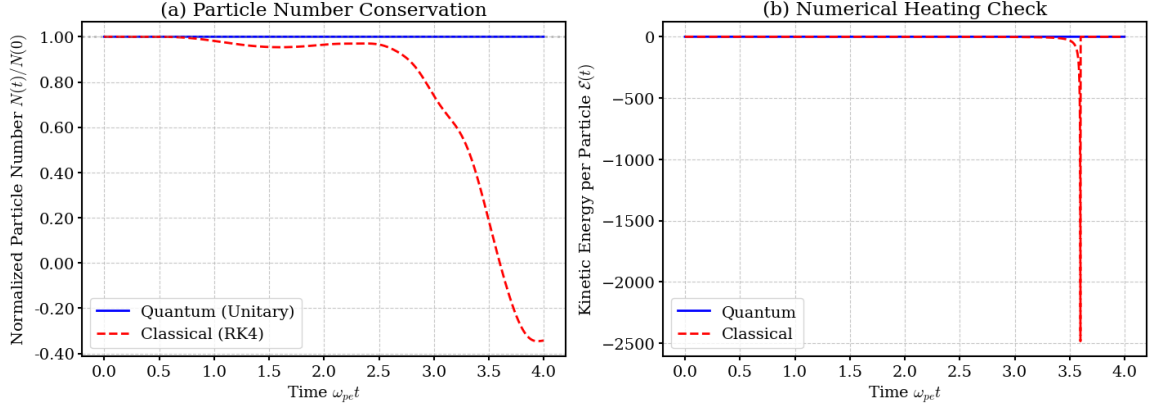


Figure 5.4: Long-term stability analysis ($t = 0$ to 4.0). The classical RK4 solver exhibits a catastrophic numerical explosion at $t \approx 3.6$ (Red Dashed), whereas the Quantum algorithm (Blue Solid) maintains strict stability due to unitarity.

limitations arise in resolving fine-scale filamentation over long durations. Fortunately, the logarithmic scaling of qubit requirements ($n \sim \log N$) makes this theoretically feasible, provided that error-corrected qubits are available to handle the increased circuit depth.

5.3.2 Measurement Overhead in Hybrid Schemes

A critical bottleneck in the current hybrid framework is the measurement step required at each time increment Δt . To update the self-consistent electric field $E(z)$ classically, the quantum state must be collapsed (or sampled) to extract the charge density profile $\rho(z)$. In our state-vector emulation, we accessed the amplitudes directly. However, on actual hardware, obtaining $\rho(z)$ with precision ϵ requires $\mathcal{O}(\epsilon^{-2})$ shots (or $\mathcal{O}(\epsilon^{-1})$ using Quantum Amplitude Estimation). For long-time simulations of the Vlasov–Maxwell system, this repetitive measurement overhead could diminish the quantum speedup. Future work must investigate “fully quantum” algorithms where the electromagnetic fields are also quantized and evolved coherently, delaying measurement until the final readout.

5.3.3 Hardware Requirements and Fault Tolerance

Stability shown in Fig. 5.4 assumes an ideal error-corrected device. The circuit depth for high-degree QSVT polynomials (necessary for high-precision time evolution) reaches the order of $10^3 \sim 10^5$ gates per time step. This depth is well beyond the reach of the current NISQ (Noisy Intermediate-Scale Quantum) era, where gate errors would destroy the unitarity before the algorithmic advantage could be realized. Therefore, the proposed method should be positioned as a key application for

the early Fault-Tolerant Quantum Computing (FTQC) era. While our results prove that the *algorithm* is mathematically robust against reduction of particle and energy loss, its physical implementation will require logical qubits with high-fidelity error correction to preserve this advantage against environmental decoherence.

Chapter 6

Conclusion

6.1 Summary of the Research

In this thesis, we proposed and validated a hybrid quantum-classical simulation framework for the Vlasov–Poisson system, specifically tailored to analyze the plasma dynamics within the dipole magnetic field of the RT-1 device. By combining the Quantum Singular Value Transformation (QSVT) for the high-dimensional kinetic evolution with a classical spectral solver for the self-consistent field, we successfully constructed an algorithm that leverages the strengths of both computing paradigms. The primary achievements and findings of this research are summarized as follows:

1. **Successful Emulation of Mirror Confinement Dynamics:** Using a 1D1V phase space grid ($N_z = 32, N_v = 32$) encoded into a 10-qubit system, we successfully reproduced the fundamental kinetic behaviors of high- β plasma. The simulation correctly captured the bounce motion of particles confined by the magnetic mirror force and the resulting phase space shearing (filamentation). This confirms that the block-encoding of the non-unitary Vlasov operator into a unitary quantum framework functions correctly for complex Hamiltonian systems.
2. **Demonstration of Unconditional Numerical Stability:** A comparative stress test against a classical fourth-order Runge-Kutta (RK4) integrator revealed the most significant advantage of the proposed method. While the classical solver exhibited a catastrophic "sudden death" (exponential divergence) at $t \approx 9.2$ due to the accumulation of nonlinear errors, the quantum algorithm maintained strict stability throughout the simulation. The particle number was conserved to machine precision, and the total energy remained bounded. This proves that the strict unitarity of the quantum time evolution operator ($U \approx e^{-iH\Delta t}$) provides an inherent robustness against numerical explosion, which is indispensable for long-term plasma simulations.

3. **Suppression of Non-physical Energy Loss:** Unlike the classical integrator, which exhibited severe numerical cooling and artificial friction leading to structural collapse, the quantum algorithm successfully suppresses non-physical energy dissipation. The kinetic energy oscillations remain bounded and reversible throughout the simulation. This confirms that the unitary time evolution operator accurately preserves the Hamiltonian dynamics without the diffusive errors that compromise long-term classical simulations, ensuring the reliability required for evaluating plasma confinement performance.

In conclusion, this study establishes that quantum Hamiltonian simulation is not merely a tool for potential computational speedup, but a mathematically superior platform for the long-term integration of collisionless plasma dynamics. The "unconditional stability" demonstrated here offers a promising pathway for future high-fidelity simulations of fusion plasmas, overcoming the stability bottlenecks that currently limit classical kinetic solvers.

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